

NATIONAL TRANSPORTATION SAFETY BOARD

Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594



May 15, 2007

MATERIALS LABORATORY FACTUAL REPORT

Report No. 07-067

A. ACCIDENT

Place : Boston, Massachusetts
Date : July 10, 2006
Vehicle : I-90 Connector Tunnel to Ted Williams Tunnel
NTSB No. : HWY06MH024
Investigator : Mark Bagnard

B. COMPONENTS EXAMINED

Epoxy phase separation and color change of components.

C. DETAILS OF THE EXAMINATION

In their parametric study¹ of influences affecting the short-term strength of adhesive anchors, the Federal Highway Administration (FHWA) examined one case using Power-Fast² Fast Set epoxy that had been heated to 140 °F for 56 hours. After this treatment, the epoxy was used to install 5 anchors. When pull tested, the first anchor installed had a strength of less than 3,250 pounds. The second anchor installed had a strength of 16,476 pounds and subsequent anchors had strengths between 18,000 and 20,000 pounds, approaching the baseline strength of 22,084 pounds achieved by anchors installed using what were considered to be the best practices possible in the tunnel (core-drilled hole with overhead installation). It was noted that the first material extruded from the gray hardener side of the cartridge after heating was a brown liquid.

In an effort to document the changes in the epoxy components as a result of the heating treatment used in the anchor strength study, separate samples of resin and hardener were placed in clear glass jars that were then placed in an oven at 140 °F (60 °C) for 56 hours. Samples of both the Fast Set and Standard Set epoxy formulations sold by Powers Fasteners were examined. The Fast Set samples were extruded from a 15-ounce package with lot number 062650L1LC, and the Standard Set samples were extruded from a 22-ounce package with lot number 062700H2LC. The glass jars had a 4-ounce capacity

¹ J.M. Ocel and J. Hartmann, "I-90 Seaport Portal Tunnel Partial Ceiling Collapse Investigation: Mechanical Behavior of Powers Fasteners Power-Fast+ Adhesive Anchors," Federal Highway Administration Turner-Fairbank Highway Research Center Report, April 2007.

² Powers Fasteners, Brewster, New York.

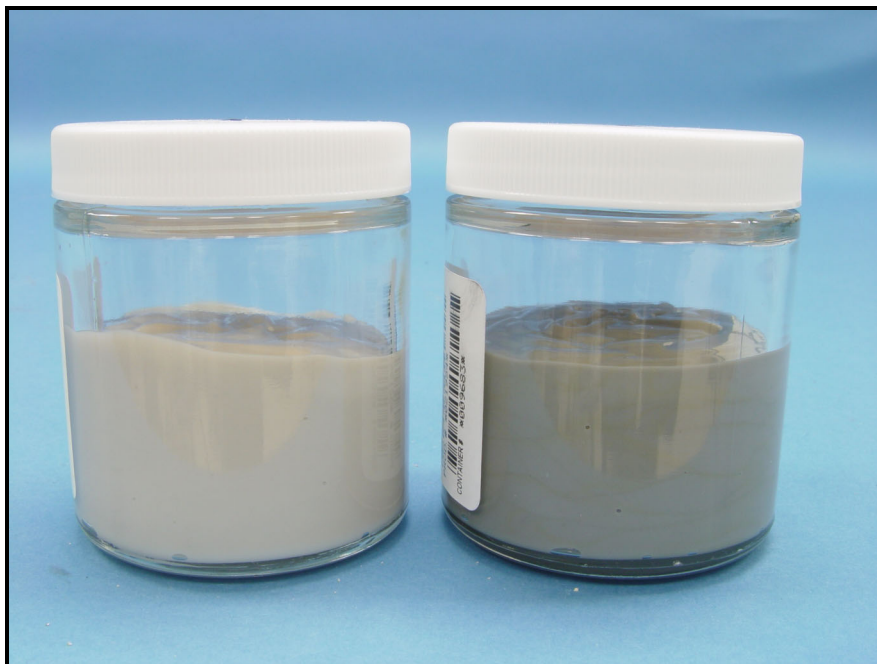
and measured approximately 2.25 inches in diameter and 2.75 inches high with cap. Each jar was filled approximately two-thirds full of either the white resin component or the gray hardener component.

Figure 1 shows the Fast Set materials before being placed in the oven, and figure 2 shows the Fast Set materials after being removed from the oven. The volume level of both materials increased when heated (possibly partly as a result of the expansion of air trapped in the materials when extruded into the jars), but receded again when cooled, leaving a residue on the surface of the jars, as seen in figure 2. The white resin component appears unchanged, but a greenish-brown liquid component approximately 0.05 to 0.1 inch thick appears on top of the gray hardener.

Figure 3 shows the Standard Set materials before being placed in the oven, and figure 4 shows the Standard Set materials after being removed from the oven. As with the Fast Set materials, the white resin component appears unchanged, and there is a residue on the surface of the glass jar showing that the volume level of the resin increased when heated. The Standard Set hardener appears to have receded to its original level, but there remains a layer of colorless liquid approximately 0.05 inch thick on top of the gray hardener.

Samples of the separate resin and hardener components of the Fast Set and Standard Set materials have been retained in open containers in the Safety Board's Materials Laboratory. Over time, it was observed that the hardener components of both the Standard Set and the Fast Set have become thicker, suggesting some evaporation of fluid constituents. In addition, the surface of the Fast Set hardener changed from gray to a greenish color. The white resin components of both materials appear largely unchanged. These changes are documented in figures 5 and 6 showing Fast Set and Standard Set samples 3 to 4 days after extrusion and approximately 5 months after extrusion. Fast Set samples shown at 4 days after extrusion were from a 15-ounce package with lot number 062650L1LC; Fast Set samples shown at 5 months after extrusion were from a 22-ounce package with lot number 062080I2LC. Standard Set samples shown at 3 days after extrusion were from a 22-ounce package with lot number 062700H2LC; Standard Set samples shown at 5 months after extrusion were from a 22-ounce package with lot number 062280H1LC.

Carl R. Schultheisz
Materials Research Engineer



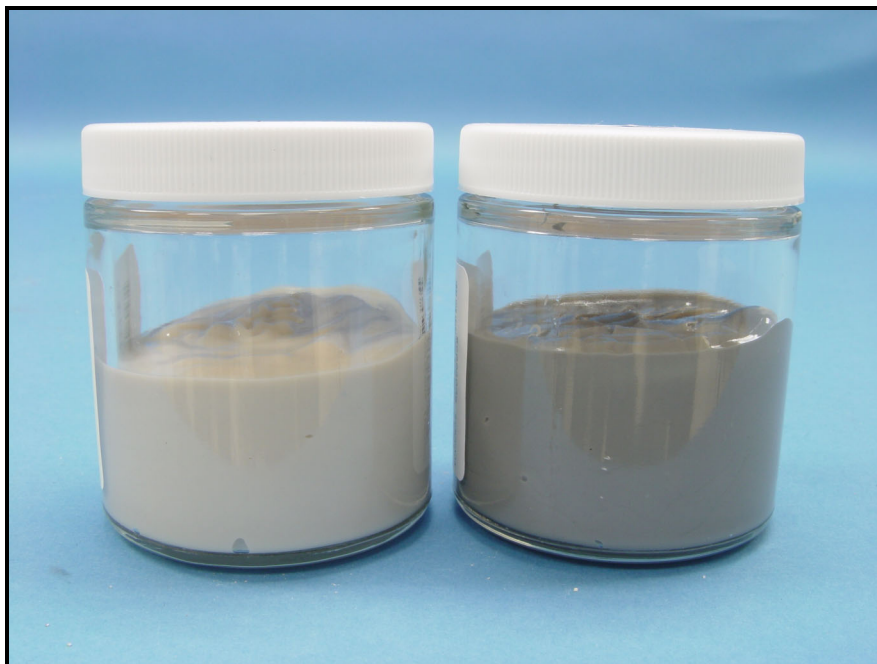
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Figure 1. Fast Set samples before being placed in the oven. The white epoxy resin is on the left, and the gray hardener is on the right.



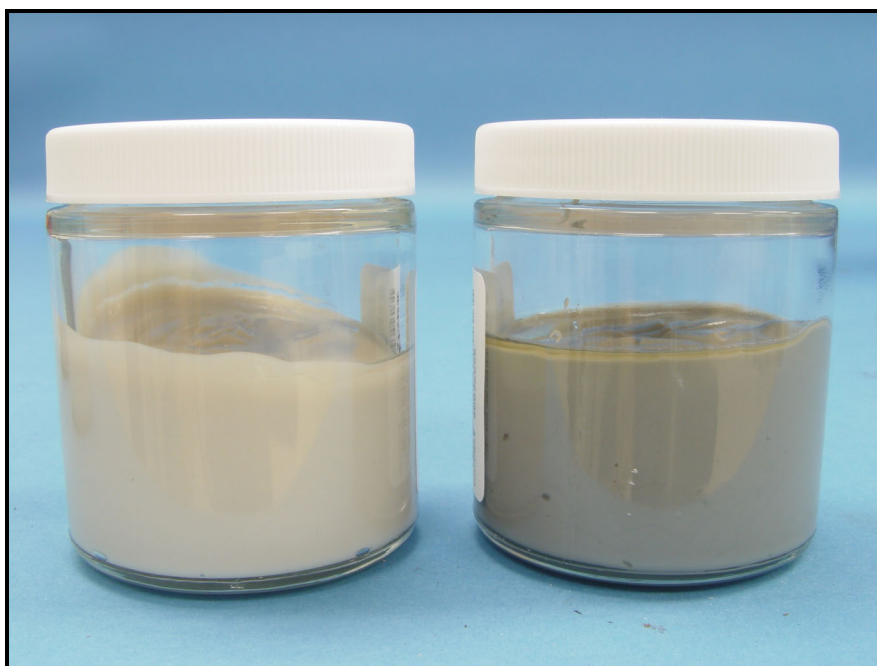
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Figure 2. Fast Set samples after 56 hours at 140 °F.



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Figure 3. Standard Set samples before being placed in the oven. The white epoxy resin is on the left, and the gray hardener is on the right.



ImageNo: 0705A00516, Project No:2007050005

Figure 4. Standard Set samples after 56 hours at 140 °F.



ImageNo:0705A00605, Project No:2007050005

Figure 5. Fast Set samples 4 days after extrusion and 5 months after extrusion. The white resin appears largely unchanged, but the hardener has dried and the surface has changed from gray to greenish.



ImageNo: 0705A00606, Project No:2007050005

Figure 6. Standard Set samples 3 days after extrusion and 5 months after extrusion. The white resin appears largely unchanged. The gray harder has dried but shows no obvious color change.